

Network System Implementation for the Marconi Law Firm, LLC.

Orlando, Florida

Jasmine Grant
JG Cyber & Cloud Solutions

Table of Contents

Project Scope	2
Network Benefits.....	3
IP Network Design Table	4
Network Configuration Process	5
User Logins and Passwords	9
Network Topology Diagram with Employee Names and Node Numbers.....	10
Capital and Operating Costs	11
Hardware and Software Specification Tables.....	11
Timesheets.....	17
Status Reports.....	19

Project Scope

Project Purpose
Why are you initiating this project for Marconi Law?
Marconi Law is starting this effort to help it realize its long-term goal of continuing to expand and create money by consolidating the work of numerous offices into one. Having all of the workers on the same network will make it easier for them to talk to each other, get things done, and keep things moving. The new Local Area Network (LAN) will make it easier for people to work together and get things done. It will also make room for growth in the future.
Objectives
What are the key objectives or high-level work specifications that you will provide to Marconi Law?
Make a Local Area Network (LAN) that works for the ten people who work in Marconi Law's new main office. That is the major purpose of this project. This includes making sure that all devices can connect to the network, setting up a secure server-based environment, and correctly managing users and data. You will also need to prepare a detailed project report that includes timesheets, progress reports, a network map, and a table of IP addresses. This is to keep track of everything that happens throughout the setup and configuration.
Deliverables
What are the project deliverables needed in terms of hardware and software?
This task comes with all the gear and software you need to set up and run the network system. There are ten PCs, ten VoIP phones, two wireless access points (WAPs), one network printer, one conference phone, and one Smart TV for the workforce. RAID is also used to set up routers and servers. It has Windows Server, client operating systems, Active Directory setup, and all the additional network services you require. You'll also get a complete report on the project and a finished network plan that explains how the site functions.
Exclusions
What is not included in this project?
This task doesn't involve setting up or keeping up internet service from a third party. It doesn't include any alterations to the structure itself, like adding new wire, patching it up, or working on the electrical. The workers won't get any training for this task. If you sign a service agreement for one year after the first installation, they will take care of it and keep it up to date.
Constraints
What are the project constraints?
This project must be completed within four weeks so the network is ready before the company moves into the new office. It also needs to stay within the assigned budget and use only approved vendors for all hardware and software. The network must be designed to work within a 4,000 square foot office space while still meeting performance, reliability, and security expectations.
Success Factors
What are at least 5 key success factors that will define this project?
• Fully Operational Network – The network must be completely set up and ready to use before the firm moves into the new office so there are no disruptions to business operations.

- **Reliable Connectivity** – All devices must be properly connected and able to communicate across the network without issues.
- **Secure Data Protection** – Sensitive legal data must be stored securely using RAID and backup solutions to prevent data loss.
- **Improved Team Collaboration** – The network must allow employees to easily communicate and share information to improve teamwork and productivity.
- **Scalability for Growth** – The system should be able to support additional users and devices in the future without needing major changes.

Timeframe / Costs

How long and how much will this project take and cost, respectively?

The renovation is expected to take four weeks, which is when the firm plans to move to its new headquarters. The project will cost approximately \$48,362.59 while still fulfilling all of its hardware, software, and network needs. The budget will include the cost of the equipment, the personnel needed to install it, and the first setup. This will make sure that the system is delivered on time and is fully functioning

Network Benefits

- **Improved Communication** – With a centralized network and all employees working in one office, Marconi Law can improve communication through VoIP phones, shared systems, and instant access to information, eliminating delays caused by operating on separate floors.
- **Increased Efficiency** – Marconi Law can streamline its daily operations by providing employees with fast access to centralized data, shared network resources, and reliable connectivity, reducing time spent on manual tasks and improving overall workflow.
- **Enhanced Data Security** – The implementation of a server with RAID configuration and a dedicated backup solution will provide a secure and centralized location for sensitive legal data, ensuring data protection, controlled access, and recovery in case of system failure.
- **Scalability for Future Growth** – The new network infrastructure will allow Marconi Law to easily add more employees, devices, and services as the firm continues to grow, without requiring a complete redesign of the system.

IP Network Design Table

Subnet IP Information	
Netmask of Subnet	255.255.255.0
First IP in Subnet	192.168.30.1
Last IP in Subnet	192.168.30.254
Total Host	254

Reserved IP Addresses	
Network ID	192.168.30.0
Broadcast Address	192.168.30.255

Creating group IP ranges is an important process for ensuring efficient IP allocation for network security purposes.

Group IP Ranges	
Computers	192.168.30.10-192.168.30.29
VoIP Phones	192.168.30.30-192.168.30.49
Network Backbone	192.168.30.1-192.168.30.9
WAPs	192.168.30.50-192.168.30.59
Other	192.168.30.60-192.168.30.79

Node	Hardware Name	Static IP address
1	Marconi Computer	192.168.30.10
2	Jones Computer	192.168.30.11
3	Stark Computer	192.168.30.12
4	Drums Computer	192.168.30.13
5	Shoemaker Computer	192.168.30.14
6	Brooks Computer	192.168.30.15
7	Long Computer	192.168.30.16
8	Adams Computer	192.168.30.17
9	Smith Computer	192.168.30.18
10	Schultz Computer	192.168.30.19
11	Marconi VoIP Phone	192.168.30.30
12	Jones VoIP Phone	192.168.30.31
13	Stark VoIP Phone	192.168.30.32
14	Drums VoIP Phone	192.168.30.33
15	Shoemaker VoIP Phone	192.168.30.34
16	Brooks VoIP Phone	192.168.30.35
17	Long VoIP Phone	192.168.30.36
18	Adams VoIP Phone	192.168.30.37
19	Smith VoIP Phone	192.168.30.38
20	Schultz VoIP Phone	192.168.30.39

21	Router	192.168.30.1
22	Server	192.168.30.5
23	Switch	192.168.30.3
24	WAP 1	192.168.30.50
25	WAP 2	192.168.30.51
26	Printer	192.168.30.60
27	Smart TV	192.168.30.61
28	Conference Phone	192.168.30.62

Network Configuration Process

User Accounts	Setup and Configuration Process
Dan Marconi Phil Jones Karen Stark Evelyn Schultz Bill Shoemaker Mike Drums Lisa Brooks Terry Long Samuel Adams Brian Smith	Step 1: Launch Windows Server 2025 VM. Step 2: Log into the server using your Administrator account. Step 3: Open Server Manager. Step 4: Access Active Directory Users and Computers - In Server Manager, click on Tools in the top right corner. - From the drop-down menu, select Active Directory Users and Computers. Step 5: Navigate to the Users container - In the left pane, expand your domain name. - Click on the Users folder. Step 6: Initiate New User Creation - Right-click on the Users container. - From the context menu, select New > User. Step 7: Enter User Information (Part 1) - In the New Object – User dialog box, enter the following: - First name: Dan - Last name: Marconi - User logon name: dmarconi - Click Next. Step 8: Set User Password and Options - Enter a password in both the Password and Confirm password fields. - Uncheck User must change password at next logon. - Click Next, then click Finish. Step 9: Repeat the user creation process

	a. Follow the same steps to create the remaining users: i. Phil Jones ii. Karen Stark iii. Evelyn Schultz iv. Bill Shoemaker v. Mike Drums vi. Lisa Brooks vii. Terry Long viii. Samuel Adams ix. Brian Smith Step 10: Verify user accounts a. Confirm that all users appear in the Users container inside Active Directory Users and Computers.
Groups	Setup and Configuration Process
_Accounting _Attorneys _Office Coordinators	Step 1: Open Active Directory Users and Computers a. In Server Manager, click Tools. b. Select Active Directory Users and Computers. Step 2: Create a new Organizational Unit (OU) for groups a. In the left pane, right-click your domain name. b. Select New > Organizational Unit. c. Name the OU: Groups. d. Click OK. Step 3: Navigate to the Groups OU a. In the left pane, click on the newly created Groups folder. Step 4: Create the _Accounting group a. Right-click inside the Groups OU. b. Select New > Group. c. Enter the group name: _Accounting. d. Set Group scope to Global. e. Set Group type to Security. f. Click OK. Step 5: Create the _Attorneys group a. Right-click inside the Groups OU. b. Select New > Group. c. Enter the group name: _Attorneys. d. Set Group scope to Global. e. Set Group type to Security. f. Click OK.

	Step 6: Create the _Office Coordinators group - Right-click inside the Groups OU. - Select New > Group. - Enter the group name: _Office Coordinators. - Set Group scope to Global. - Set Group type to Security. - Click OK. Step 7: Add users to the _Accounting group - Double-click the _Accounting group. - Click the Members tab. - Click Add. - Enter: Brian Smith. - Click Check Names, then click OK. - Click Apply, then OK. Step 8: Add users to the _Office Coordinators group - Double-click the _Office Coordinators group. - Click the Members tab. - Click Add. - Enter: Evelyn Schultz. - Click Check Names, then click OK. - Click Apply, then OK. Step 9: Add users to the _Attorneys group - Double-click the _Attorneys group. - Click the Members tab. - Click Add. - Enter the following users: - Dan Marconi - Phil Jones - Karen Stark - Bill Shoemaker - Mike Drums - Lisa Brooks - Terry Long - Samuel Adams - Click Check Names, then click OK. - Click Apply, then OK. Step 10: Verify group membership - Open each group. - Confirm the correct users are listed in the Members tab.

Folders and Permissions	Setup and Configuration Process
Accounting Attorneys Office Coordinators	<u>Create Folders</u> Step 1: Open File Explorer a. On the server, click the File Explorer icon on the taskbar. Step 2: Navigate to the Local Disk (C:) a. In File Explorer, click This PC. b. Double-click Local Disk (C:). Step 3: Create the main shared folder a. Right-click inside the window. b. Select New > Folder. c. Name the folder: Marconi Law. Step 4: Create subfolders inside Marconi Law a. Double-click the Marconi Law folder. b. Right-click inside the folder. c. Select New > Folder and create the following folders: i. Accounting ii. Attorneys iii. Office Coordinators <u>Set Sharing Permissions</u> Step 1: Launch Windows Server 2025 VM. Step 2: Log in using the Administrator account. Step 3: Open File Explorer. Step 4: Navigate to This PC > Local Disk (C:) > Marconi Law. Step 5: Share the Accounting folder a. Right-click the Accounting folder. b. Click Properties. c. Click the Sharing tab. d. Click Share. e. In the field, type: _Accounting f. Click Add. g. Set the permission level to Read/Write. h. Click Share. i. Click Done. Step 6: Share the Attorneys folder a. Right-click the Attorneys folder. b. Click Properties. c. Click the Sharing tab. d. Click Share. e. Type: _Attorneys f. Click Add.

	g. Set permission to Read/Write. h. Click Share, then Done. Step 7: Share the Office Coordinators folder a. Right-click the Office Coordinators folder. b. Click Properties. c. Click the Sharing tab. d. Click Share. e. Type: _Office Coordinators f. Click Add. g. Set permission to Read/Write. h. Click Share, then Done. Step 8: Verify folder sharing a. Open File Explorer. b. Click Network. c. Click your server name. d. Confirm you see: i. Accounting ii. Attorneys iii. Office Coordinators Step 9: Verify that each group only has access to its correct folder.
--	--

User Logins and Passwords

Employee Name	Job Title	Login ID	Password
Dan Marconi	President/Attorney	dmarconi	Fullsail1!
Phil Jones	Attorney	pjones	Fullsail1!
Karen Stark	Attorney	kstark	Fullsail1!
Evelyn Schultz	Office Coordinator	eshultz	Fullsail1!
Bill Shoemaker	Attorney	bshoemaker	Fullsail1!
Mike Drums	Attorney	mdrums	Fullsail1!
Lisa Brooks	Attorney	lbrooks	Fullsail1!
Terry Long	Attorney	tlong	Fullsail1!
Samuel Adams	Attorney	sadams	Fullsail1!
Brian Smith	Accounting	bsmith	Fullsail1!

Network Topology Diagram with Employee Names and Node Numbers



Capital and Operating Costs

Hardware and Software	Per Unit Cost	Quantity	Total Cost
Laptops	\$956.99	10	\$9,569.00
Router	\$2,365.99	1	\$2,365.99
Server	\$6,045.00	1	\$6,045.00
Switch	\$999.99	1	\$999.99
Server Hard Drives (Consider a RAID 5)	\$273.99	3	\$821.97
Local/Cloud Data Backup Solution	\$959.99	1	\$959.99
VoIP Phones	\$369.99	10	\$3,699.90
Conference Phone	\$469.99	1	\$469.99
WAPs	\$149.99	2	\$299.99
Printer	\$299.99	1	\$299.99
Smart TV	\$399.99	1	\$399.99
Windows Server 2025	\$879.99	1	\$879.99
Windows 11	\$154.99	10	\$1,549.90
Labor	\$125	40 hrs	\$5,000
Annual ongoing network maintenance and support	\$15,000	1 yr	\$15,000
GRAND TOTAL	-	-	\$48,362.59

Hardware and Software Specification Tables

Laptops

Brand/Manufacturer	HP
Product Line	HP 250R
Model	G10
Processor/Clock Speed	Intel Core 5 120U (up to 5.0 GHz)
Memory/Speed	16 GB DDR4
Hard Drive/Capacity	512 GB SSD
Embedded Security	TPM 2.0
Display/Size/Resolution	15.6" / 1920 × 1080 (Full HD, 60 Hz)
Networking	Wi-Fi 6, Bluetooth, Gigabit Ethernet
Operating System	Windows 11 Pro
Cache	12 MB Intel Smart Cache
Page URL	https://www.cdw.com/product/hp-250r-g10-15.6-notebook-full-hd-60-hz-intel-core-5-120u-16-gb/8403180

Server

Brand/Manufacturer	Hewlett Packard Enterprise (HPE)
Product Line	ProLiant
Model	DL320 Gen11
Hard Drive/Capacity	RAID 5 (3 × 1.2TB SAS drives)
Processor	Intel Xeon Bronze 3508U
Memory/Speed	16 GB DDR5
Storage Controller	HPE Smart Array Controller (RAID supported)
Monitor	Not included
Ethernet	Dual Gigabit Ethernet ports
Operating System	Windows Server 2025
Networking	Ethernet LAN
Expansion Bays	2.5" SFF bays
Expansion Slots	PCIe expansion slots
Page URL	https://www.cdw.com/product/hpe-proliant-dl320-gen11-rack-mountable-xeon-bronze-3508u-2.1-ghz-16-gb/8412221

Server Hard Drives

Brand/Manufacturer	Seagate
Product Line	Exos 10E2400
Storage Type	HDD
Hard Drive Type	Enterprise
Form Factor	2.5" (SFF)
Interface	SAS 12Gb/s
Capacity	1.2TB
Data Transfer Rate (DTR)	12 Gb/s
Page URL	https://www.cdw.com/product/seagate-exos-10e2400-st1200mm0129-hybrid-hard-drive-1.2-tb-sas-12gb-s/5009289

Router

Brand/Manufacturer	Cisco/ Cisco Systems
Product Line	Cisco Catalyst
Model	8200-1N-4T
RAM	8 GB
Flash Memory	8 GB
WAN Ports Qty	4
Integrated Switch	No
Data Link Protocol	Ethernet, Gigabit Ethernet
Network/Transfer Protocol	IP, TCP/IP, MPLS, OSPF
Page URL	https://www.cdw.com/product/cisco-catalyst-8200-1n-4t-router-rack-mountable/7364597

Switch

Brand/Manufacturer	Aruba
Product Line	HPE Aruba
Model	6000 48G 4SFP Switch
RAM	4 GB
Flash Memory	16 GB
Performance	Latency (1 Gbps): 1.9 μ s, Packet buffer size: 12.38 MB, Switching capacity: 104 Gbps, Throughput: 77.3 Mpps
Capacity	ARP entries: 1024, IGMP groups: 512, IPv4 ACL entries (ingress): 256, IPv4 routes (Unicast): 512, IPv6 ACL entries (ingress): 128, IPv6 routes (Unicast): 512, MAC ACL entries (ingress): 256, MLD groups: 512, Switched virtual interfaces (SVIs): 16
Ports Qty*	48
Page URL	https://www.cdw.com/product/hpe-networking-6000-48g-4sfp-switch-switch-48-ports-managed-rack-mo/6768074

**Switch needs to be capable of hard wiring all hardware (nodes or endpoints) for your LAN.*

Local/Cloud Data Backup Solution

Brand/Manufacturer	Synology
Capacity	Depends on installed drives (supports RAID; scalable storage up to ~40TB+ depending on drives)
Speed	Gigabit Ethernet (expandable for higher throughput)
Encryption/Security	AES-256 encryption, user authentication, access control
Compliance	Supports business data protection standards (backup, recovery, encryption)
On Premises/Offsite	On-premises with optional cloud sync (hybrid backup)
Backup Strategy	RAID redundancy + scheduled backups + cloud replication
Page URL	https://www.cdw.com/product/synology-4-bay-rackstation/7064333?pfm=srh

Printer

Brand/Manufacturer	HP
Product Line	LaserJet Pro
Model	MFP M428fdw
Duty Capacity	High volume
Printing Output Type	Monochrome
Processor	1200 MHz
RAM	512MB
Networking	Ethernet, Wi-Fi
Page URL	https://www.cdw.com/product/hp-laserjet-pro-3101fdw-wired-wireless-laser-multifunction-printer-mono/7198794?pfm=srh

Wireless Access Points (WAPs)

Brand/Manufacturer	HPE Aruba
Product Line	Instant On
Model	AP22
Antenna	Internal
Capacity	Multiple users
Networking	Wi-Fi 6
Interface	Ethernet
Power	PoE
Page URL	https://www.cdw.com/product/hpe-networking-instant-on-ap22-us-wireless-access-point-bluetooth-wi/6195848?pfm=srh

Smart TV

Brand/Manufacturer	Samsung
Model	UN58U8000FF
Series	U8000F Series
Video Interface	HDMI
HDMI Ports Qty	3
Diagonal Size	58" (57.5" viewable)
Resolution	3840 x 2160 (4K UHD)
Display Format	LED-backlit LCD
Viewing Angle	Wide Viewing Angle
Digital Audio Format	Dolby Digital Plus
USB Ports Qty	2
Wi-Fi Protocol	Wi-Fi 5 (802.11ac)
Page URL	https://www.cdw.com/product/samsung-un58u8000ff-u8000f-series-58-class-57.5-viewable-led/8370698

Conference Phone

Brand/Manufacturer	Poly
Product Line	Trio
Model	8300
Type	Conference VoIP
Call Services	VoIP
Speakerphone	Yes
Features	HD voice
Phone connections	Ethernet
Page URL	https://www.cdw.com/product/samsung-un58u8000ff-u8000f-series-58-class-57.5-viewable-led-backlit/8370698?pfm=srh

Server Software

Brand/Manufacturer	Windows
Product Line	Windows Server 2025
Version	Standard
Licensing	16 cores
Page URL	https://www.cdw.com/product/microsoft-windows-server-standard-edition-license-16-cores/4878963?pfm=srh

Client Computer Software

Brand/Manufacturer	Windows
Product Line	Windows 11
Version	11
Licensing	1
Page URL	https://www.cdw.com/product/windows-11-pro-license-1-license/6729569?pfm=srh

VoIP Phones

Brand/Manufacturer	Cisco
Product Line	IP Phone
Model	8841
Type	VoIP
Conference Call Capability	Yes
Call Services	VoIP
Speakerphone	Yes
Additional Functions	Hold, transfer, voicemail
Features	HD voice
VoIP Protocols	SIP
IP Address Assignment	DHCP
Network Protocols	TCP/IP
PoE Support	Yes
Voice Features	Wideband audio
Page URL	https://www.cdw.com/product/cisco-ip-phone-8841-voip-phone/5329935?pfm=srh

Timesheets

Milestone	Activity Description (be descriptive)	Start Time	End Time	Total Activity Time	Activity Date
Milestone 1	Based on the Marconi Law Firm case study, I looked into the hardware needs and chose the right equipment (laptops, server, router, switch) using CDW.	2:00 PM	6:00 PM	4hrs	Apr 7, 2026
Milestone 1	Finished entire hardware and software specification tables, which included setting up a RAID 5 server and checking that all parts worked together.	3:00 PM	7:00 PM	4 hrs	Apr 8, 2026
Milestone 1	Created and improved the network topology diagram, putting backbone devices in the mechanical room and making sure all network devices were there.	2:00 PM	5:30 PM	3.5 hrs	Apr 9, 2026
Milestone 2	Installed Active Directory Domain Services and began configuring the server as a domain controller	10:00 AM	2:00 PM	4hrs	Apr 15, 2026
Milestone 2	Assigned static IP addresses to all network devices and completed the IP Network Design Table	10:00 AM	2:00 PM	4 hrs	Apr 16, 2026
Milestone 2	Completed the project scope and network objectives and updated the network topology diagram with correct labels and node numbers	10:00 AM	1:30PM	3.5hrs	Apr 17, 2026
Milestone 3	Created and configured all user accounts in Active Directory for the Marconi Law employees and verified	10:00 AM	2:00 PM	4hrs	Apr 21, 2026

	that each account was able to log in correctly.				
Milestone 3	Created security groups, configured shared folders, and applied sharing permissions to control access based on employee job roles and departments.	10:00 AM	2:00 PM	4 hrs	Apr 23, 2026
Milestone 3	Tested mapped network drives and completed the final recording of the video demonstration showing user access, folder permissions, and domain controller configuration.	10:00 AM	1:30PM	3.5hrs	Apr 24, 2026
Milestone 4	Finalized and professionally formatted the full project report, corrected formatting issues, removed instructional text, and updated the table of contents.	10:00 AM	1:00 PM	3 hrs	Apr 28, 2026
Milestone 4	Created the PowerPoint pitch presentation for Marconi Law and organized slides covering network scope, costs, specifications, security, and benefits.	2:00 PM	6:00 PM	4hrs	Apr 29, 2026
Milestone 4	Recorded and reviewed the final client pitch presentation video, including narration, transitions, and presentation timing to stay under 10 minutes.	11:00 AM	2:30PM	3.5hrs	Apr 30, 2026
TOTAL HOURS				45 HRS	

Status Reports

Accomplished this week

Milestone 1: Did research and chose all the network hardware needed using CDW, making sure it would operate in a legal office with 10 users.
Milestone 1: Made extensive tables of hardware and software requirements, including how to set up a server with RAID 5 storage and the right licenses.
Milestone 1: Made the first network topology diagram and put the network backbone (router, switch, and server) in the right spot in the mechanical room.
Milestone 2: Installed and configured Active Directory Domain Services on the Windows Server and promoted it to a domain controller for the network.
Milestone 2: Created and assigned static IP addresses for all devices and completed the IP Network Design Table to match the network layout.
Milestone 2: Completed the project scope and network objectives, making sure they aligned with the business case and clearly defined the project requirements.
Milestone 3: Created organizational units (OUs), user accounts, and security groups in Active Directory to support employee management and access control.
Milestone 3: Configured employee usernames and passwords, mapped shared folders to the client workstation, and tested employee access to verify that users could only access the folders appropriate for their department.
Milestone 3: Recorded and presented a full video walkthrough of the network setup, including employee account creation, shared folder permissions, and client-side access testing to demonstrate that the system was functioning correctly.
Milestone 4: Completed and properly structured the whole project report, which included fixing tables, updating formatting, and getting rid of instructional material.
Milestone 4: Made a professional pitch presentation for Marconi Law that explained the network design, hardware choices, security issues, and commercial benefits.
Milestone 4: Watched and recorded the final presentation film, which included a voiceover detailing the finished network solution and the project's results.

Planned for next week

Coming up for milestone 2: Improve and finish the network topology diagram by making sure that all devices, such as VoIP phones, WAPs, and endpoints, are connected correctly.
--

Coming up for milestone 2: Set up Active Directory on the server and make it a domain controller. Then, test the network.
Coming up for milestone 2: Establish a table of static IP addresses for all network devices and finish the project's goals and scope.
Coming up for milestone 3: Create and configure all user accounts in Active Directory and make sure each employee can log in correctly using their assigned credentials.
Coming up for milestone 3: Set up security groups and shared folders, then apply permissions to control access based on employee roles.
Coming up for milestone 3: Document the full configuration process step by step in the project report and begin preparing the video demonstration of the domain controller setup.
Coming up for milestone 4: Finalize the project report formatting and ensure all sections are professionally completed.
Coming up for milestone 4: Complete and record the final pitch presentation for Marconi Law.
Coming up for milestone 4: Review the full project for accuracy, organization, and professionalism before final submission.

Comments for each milestone

<u>Milestone 1:</u> What problems did you run into? One issue I had was picking the right hardware that met the project's needs and worked well in the actual world. At first, I picked a tower server over a rack-mounted server, which was not the suggested arrangement. I was also a little confused about how to make sure the server hard drives worked with the server, especially when it came to RAID 5 needs. How did you fix them? I solved these problems by going over the project instructions again and changing the gear I chose to better resemble a professional network configuration. I converted to a rack-mounted server and made sure that the hard drives were the right size (2.5" SAS) and would work with the server. I also reviewed all of the product specifications on CDW again to make sure they were all proper. Is there anything that you did not complete that you now have to push off into the next milestone? I was able to do most of the Milestone 1 tasks, but I still need to finish and improve the network topology design to make sure that all the devices are in the right position and linked properly. I will finish this by the next milestone. Any other comments?

This milestone taught me a lot about how vital it is to plan ahead and make sure everything function together while building a network. I also learnt how each part of the system works together, notably the switch, router, and server. I feel better about moving on to the next step and making my network design better.

Milestone 2:

What problems did you run into?

Setting up Active Directory and making sure the domain manager was set up correctly was one of the hardest things I had to do. There were also times when the Proxmox virtual lab wouldn't start, which made things take longer. I also had to check that the project's goals and scope were the same as what was in the business case and that the static IP addresses were set up correctly.

How did you fix them?

I was able to fix this by going back and reading the directions again and putting things together one step at a time. To fix the Proxmox problem, I restarted the session and waited for the computer to start up again. I also checked my IP address database one last time to make sure it was all correct. I also changed the project's scope to make sure that the goals, objectives, and deliverables were clear.

Is there anything that you did not complete that you now have to push off into the next milestone?

I am done with Milestone 2, but I still need to test the domain controller to make sure that all of the client computers can connect. This is what I want to do next.

Any other comments?

This big step helped me understand how better planning and setting up a network work together. It was harder than the last one, but it helped me learn how the overall network works.

Milestone 3:

What problems did you run into?

I had a hard time making sure that everyone who used the shared drive could only access the folders they were supposed to. Some people could open folders that they shouldn't have been able to at initially. It was also hard for me to tell the difference between sharing permissions and Windows Server security rights. I also had problems using the mapped network disks on the client machine because some of the Windows 11 user accounts weren't set up yet.

How did you fix them?

To fix these problems, I reviewed the class videos again and double-checked the permissions for each folder. I checked that each employee was in the correct Active Directory security group and deleted any permission records that gave users too much authority. Before I made the video example, I entered onto the client computer as Brian Smith, Dan Marconi, and Evelyn Schultz. This manner, their profiles would be ready to utilize right away. I looked over the folders more than occasions to make sure that each user could only go to the folders that were meant for their section.

Is there anything that you did not complete that you now have to push off into the next milestone?

I didn't have to hurry to get any work done for the following goal. I did everything that needed to be done for Milestone 3. This included building user accounts, setting up groups, making shared folders and rights, testing user access, and writing a project report that outlined how to set up the system.

Any other comments?

This aim helped me understand how to use shared files and Active Directory groups together on a company network. It also taught me how crucial it is to double-check permissions to make sure that users can only see the files they are supposed to.

Completing this milestone taught me more about how to manage network users and run Windows Server.

Milestone 4:

What problems did you run into?

One of the hard things I had to do for Milestone 4 was put all the project material into a presentation that was professional, easy to follow, and still interesting. Because there was a lot of technical information, I had to make sure that the presentation made the network apparent without confusing the audience. I also had to make sure that the way the report was set up was the same all the way through.

How did you fix them?

I fixed these problems by going over the project step by step and breaking the information down into smaller, easier-to-understand parts. I also practiced my speech many times to make sure it flowed well and was on time. I made sure that the report appeared professional and consistent by carefully checking the layout, space, font sizes, and tables.

Is there anything that you did not complete that you now have to push off into the next milestone?

No. I finished all I needed to do for Milestone 4, such as writing the report, making the presentation, and filming the final project pitch video.

Any other comments?

This project taught me more about how to manage IT projects by showing me how technical setup, documentation, communication, and presentation abilities all work together. It also made me feel more sure of my ability to describe technological systems in a way that is clear and professional.